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Secure Login and Authentication

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SYSTEM CONTEXT

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Mobile and web applications store a large amount of personal and sensitive information such as user profiles, academic records, financial data, and communication records. However, many systems still rely on basic login methods that are vulnerable to security threats such as password guessing, phishing, brute-force attacks, and unauthorized access.

One of the most common problems in existing systems is weak authentication. Users often create simple passwords, reuse credentials, or fall victim to cyber threats. As a result, accounts may be compromised, leading to data loss, privacy violations, and system misuse.

To address these issues, this project proposes the development of a Secure Login and Authentication Application designed to strengthen user verification and protect sensitive information. The system will implement improved authentication mechanisms such as encrypted password storage and a One-Time Password (OTP) verification system with a 30-second cooldown timer.

The application will generate a unique login code that users must enter when logging into another system. Each OTP will only be valid for 30 seconds and will automatically refresh after the cooldown period to ensure security.

This application will help ease existing problems by:

- Preventing unauthorized access to user accounts
- Improving user privacy and data protection
- Providing a more reliable and secure login process
- Generating time-limited OTP codes every 30 seconds
- Reducing the risk of hacking and identity theft
- Demonstrating practical implementation of cybersecurity principles

The system is intended to be simple, user-friendly, and efficient while maintaining strong security measures. It can be used in schools, organizations, and small businesses that require secure login authentication.

OBJECTIVE OF THE SYSTEM

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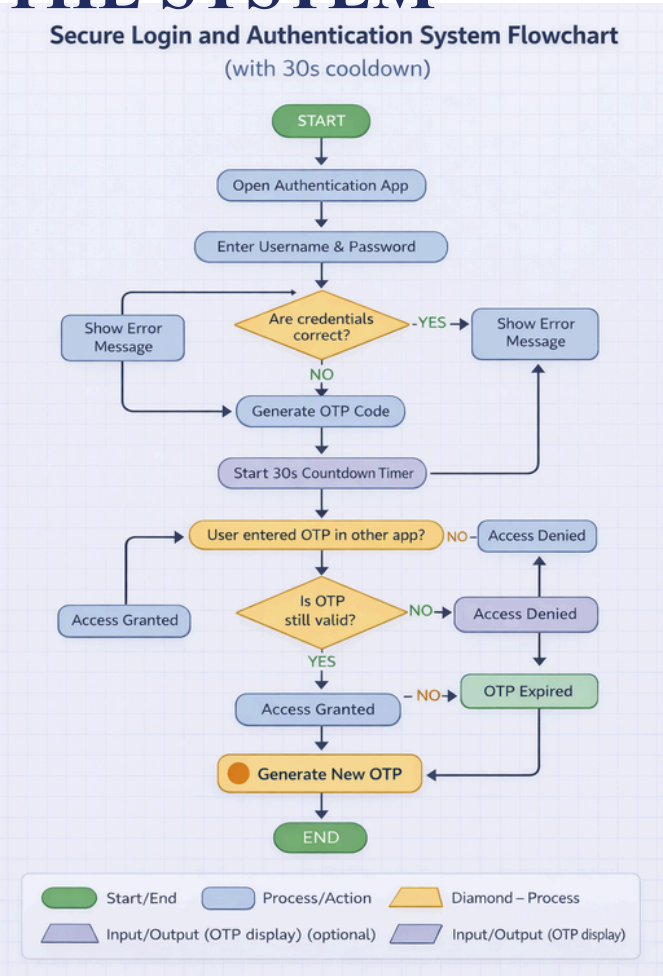
The general objective of this project is to design and develop a secure login and authentication application that generates a time-limited OTP code with a 30-second cooldown to protect user accounts and prevent unauthorized access.

Specific Objectives

The system specifically aims to:

1. Design a secure and user-friendly login interface.
2. Implement password encryption to protect user credentials.
3. Develop an OTP authentication system for additional security.
4. Generate a unique login code that refreshes every 30 seconds.
5. Display a countdown timer before generating a new OTP code.
6. Prevent reuse of expired OTP codes after the cooldown period.
7. Ensure only authorized users can access the system.
8. Improve data privacy and protection through secure authentication.
9. Demonstrate how OTP systems enhance cybersecurity in real applications.
10. Test the system for reliability, usability, and security effectiveness.

FLOWCHART OF THE SYSTEM



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UI Design

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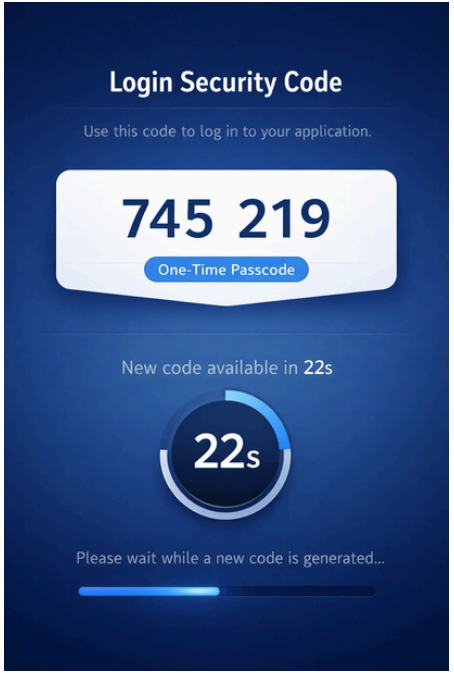

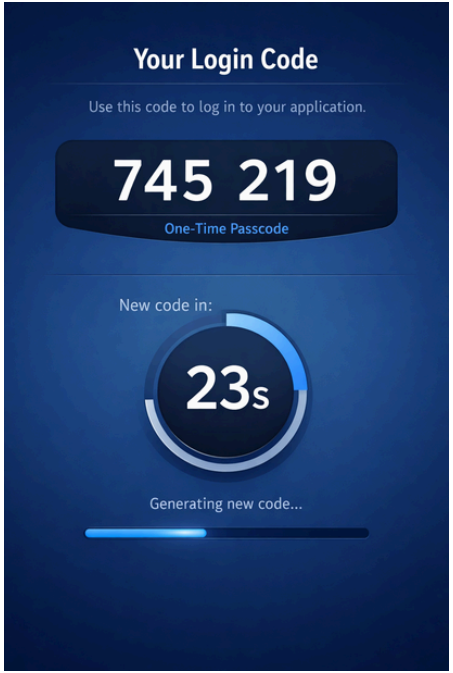

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THANK YOU!